BRIDGES IN MATHEMATICS • GRADE 3 • UNIT 4

Measurement & Fractions: Unit Fractions & Tower Heights

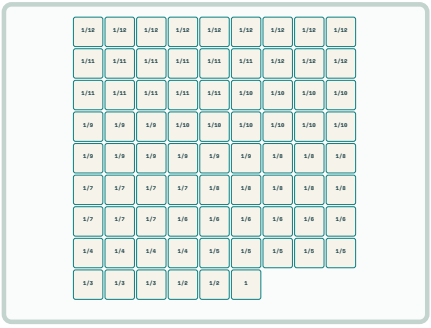
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.3.NF.A.1

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Fraction Blocks

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place one 1/4 block (21mm) and one 1/8 block (10.5mm). Notice 1/4 is twice 1/8.

Sketch Model on Paper:
Draw a fraction strip showing 1/4 shaded out of 4 equal parts.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	FDISP	Fraction Studio	
1.	·	X: .	Initiate fraction en
2.	1	X: 1/	Numerator 1 entered
3.	·	X: 1/	Denominator separato
4.	6	X: 1/6	Denominator 6 -> 1/6

EXPECTED LCD X: 1/6 (0.1667)

1 **Enter unit fraction 1/5: [.] [1] [.] [5]**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Enter unit fraction 1/12: [.] [1] [.] [1][2]**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Add two 1/3 blocks: 1/3 + 1/3.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **How many 1/8 blocks make 1/2?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 4

Measurement & Fractions: Unit Fractions & Tower Heights • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Fraction Studio Proportional Towers

CAD Blueprint: fraction_blocks

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/fraction_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Precision Proportional Fraction Blocks (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Enter unit fraction 1/5: [.] [1] [.] [5]** Standard: CCSS.MATH.3.NF.A.1	1/5	[.] [1] [.] [5]
2	**Enter unit fraction 1/12: [.] [1] [.] [1][2]** Standard: CCSS.MATH.3.NF.A.1	1/12	[.] [1] [.] [1][2]
3	**Add two 1/3 blocks: 1/3 + 1/3.** Standard: CCSS.MATH.3.NF.A.1	2/3	[.] [1] [.] [3] [ENTER] [.] [1] [.] [3
4	**How many 1/8 blocks make 1/2?** Standard: CCSS.MATH.3.NF.A.1	4	[.] [1] [.] [2] [ENTER] [.] [1] [.] [8

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Larger denominator means smaller unit piece height.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.